

# **Executive Business Report**

## **End-to-End Sales Forecasting & Demand Intelligence System**

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### **Executive Summary**

This project was developed to help the company make better inventory and sales planning decisions using historical sales data. By analyzing four years of retail sales transactions, the system identifies sales trends, predicts future demand, detects unusual sales patterns, and groups products based on their demand behavior. These insights help reduce stock shortages, avoid unnecessary inventory costs, and improve overall business planning. Among the forecasting methods tested, the XGBoost model delivered the most accurate predictions and is recommended for future business use. Overall, this system provides a practical decision-support tool that can help improve operational efficiency and customer satisfaction.

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### **1. Key Findings from Data Analysis and Forecasting**

#### **Sales Analysis**

- Technology products generated the highest overall revenue during the study period, making them the strongest-performing product category.

- The West region consistently recorded strong sales compared with other regions, indicating stable customer demand.
- Sales showed clear seasonal patterns, with noticeable increases during holiday and festive shopping periods.
- The average shipping time remained consistent across regions, suggesting an efficient order fulfillment process.

### Forecasting Performance

Three forecasting approaches were evaluated to estimate future monthly sales.

| Model   | MAE      | RMSE     | MAPE          |
|---------|----------|----------|---------------|
| SARIMA  | 6338.47  | 6377.25  | 36.79%        |
| Prophet | 12050.66 | 12505.38 | Not available |
| XGBoost | 2545.55  | 2722.47  | 14.80%        |

Based on these results, **XGBoost** provided the most accurate forecasts and is recommended for operational use.

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## 2. Three-Month Sales Forecast

Using the XGBoost forecasting model, sales are expected to remain stable over the next three months with normal seasonal fluctuations.

The forecast suggests:

- Customer demand is expected to remain strong for Technology products.
- The West region is likely to continue generating the highest sales.

- No significant decline in overall sales is expected during the forecast period.
- Normal month-to-month variations should be anticipated due to seasonal buying patterns.

The forecast confidence indicates that actual sales are expected to remain close to the predicted values under normal business conditions.

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### **3. Top Three Sales Anomalies and Their Likely Causes**

During the analysis, several unusual sales weeks were identified.

#### **Anomaly 1 – Sudden Sales Spike**

##### **Likely Cause:**

Seasonal promotions or festive shopping events resulted in unusually high customer purchases.

#### **Anomaly 2 – Sharp Sales Drop**

##### **Likely Cause:**

Possible inventory shortages, supply chain delays, or temporary reductions in customer demand.

#### **Anomaly 3 – Unexpected High Sales Week**

##### **Likely Cause:**

Special marketing campaigns, discounts, or bulk customer purchases increased sales beyond normal levels.

Monitoring these unusual events allows management to prepare inventory and staffing more effectively.

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### **4. Product Demand Segmentation and Stocking Strategy**

Products were grouped into demand-based segments to improve inventory planning.

## **High Volume, Stable Demand**

### **Characteristics**

- Consistent customer demand
- Reliable revenue generation

### **Recommended Strategy**

Maintain sufficient inventory throughout the year to prevent stock shortages.

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## **Growing Demand**

### **Characteristics**

- Sales increasing steadily
- Strong future growth potential

### **Recommended Strategy**

Gradually increase inventory and monitor demand closely.

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## **Low Volume, High Volatility**

### **Characteristics**

- Irregular sales patterns
- Difficult to predict demand

### **Recommended Strategy**

Maintain limited inventory and review stock frequently.

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## **Declining Demand**

### **Characteristics**

- Falling sales trend
- Lower customer demand

### **Recommended Strategy**

Reduce inventory levels and use promotional campaigns to clear existing stock.

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## **5. Business Recommendations**

### **Recommendation 1**

Increase inventory allocation for Technology products and the West region because they consistently generated the highest sales throughout the analysis period.

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### **Recommendation 2**

Adopt the XGBoost forecasting model for monthly inventory planning since it achieved the lowest forecasting errors (MAE: **2545.55**, RMSE: **2722.47**, MAPE: **14.80%**) among all evaluated models.

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### **Recommendation 3**

Establish a weekly sales monitoring process to identify unusual spikes or drops in sales. Early identification of these events will

allow faster responses to promotions, supply disruptions, or changing customer demand.

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## **6. Risk and Limitation**

This forecasting system is based entirely on historical sales data. It does not consider external factors such as inflation, competitor activities, economic conditions, weather, marketing campaigns, or unexpected events. As a result, future forecasts may differ if major business conditions change. The model should therefore be updated regularly with new sales data and reviewed alongside business knowledge before making important inventory decisions.

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## **Conclusion**

This project demonstrates how historical sales data can be transformed into practical business insights for inventory management and strategic planning. By combining sales trend analysis, forecasting, anomaly detection, and product demand segmentation, the system provides a structured approach to improving business decisions. The XGBoost model delivered the highest forecasting accuracy and is recommended for future implementation. With regular updates and continuous monitoring, this system can help reduce inventory costs, improve product availability, and support long-term business growth.